

Abdullah Al Masud

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ACADEMIC BACKGROUND

Ph.D. in Mechanical Engineering <i>Texas Tech University, USA (Advisor: Dr. Jingfei Liu)</i>	Summer 2021-Spring 2024 CGPA 3.91 out of 4.00
M.Sc. in Mechanical Engineering <i>Texas Tech University, USA</i>	Fall 2017-Spring 2021 CGPA 3.90 out of 4.00
B.Sc. in Mechanical Engineering <i>Islamic University of Technology (IUT), Bangladesh</i>	December 2015 CGPA 3.86 out of 4.00

Technical Skills

Ultrasound Imaging and Signal Processing: Elastography, Doppler Imaging, Total Focusing Method (TFM), Full wave Inversion (FWI), Reverse Time Migration (RTM), Phased Array Ultrasound Testing (PAUT), Beamforming, Image Reconstruction, Phase Aberration Correction, Spatiotemporal Signal Analysis, Resonance Ultrasound Spectroscopy (RUS)

Machine Learning and Artificial Intelligence: Supervised and Unsupervised Learning, Deep Learning Algorithms(CNN, RNN, LSTM, Autoencoders, U-Net, Transformer-based models), Physics Informed Neural Network (PINN)

Frameworks: TensorFlow, PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, SciPy, Matplotlib

Modeling and Simulation: Finite Element Modeling (ANSYS, Abaqus, FEBio), ultrasound field simulation (K-Wave, Field II, Focus), acoustic scattering, and wave propagation analysis

Instrumentation and Experimental Systems: Verasonics Vantage 256, Thorlabs OCT, Digital Image Correlation (DIC), Hydrophones, Function Generators, Oscilloscopes, Data Acquisition Systems

Programming and Software Tools: Python, MATLAB, C/C++, R, SQL, SolidWorks, Simulink, JMP, LabVIEW, Git, PLC programming, and GD&T, Inventor

Experimental Design and Data Analysis: Design of Experiments (DOE), regression and statistical modeling, uncertainty quantification, and process optimization

Fabrication and Design: Stereolithography (SLA), Fused Deposition Modeling (FDM), precision machining, casting, SolidWorks, Inventor, GD&T, Visual Studio

Development and Quality: Kaizen, Kanban, DMAIC, FMEA, 5W, 8D, 5S, KPI, VSM, DOE

CURRENT POSITION

Research Fellow, Department of Radiology, Mayo Clinic, Rochester, Minnesota July 2024 – present

- Analyzing raw per-channel ultrasound RF data to characterize and quantify Doppler twinkling artifacts in biopsy markers (e.g., PMMA).
- Designing and implementing advanced image reconstruction pipelines, including beamforming, RF-to-IQ conversion, Doppler velocity estimation, and color/power Doppler map generation.
- Developing quantitative frameworks for twinkling analysis to enhance marker conspicuity and diagnostic reliability in clinical ultrasound and MRI.
- Performing MRI image processing to evaluate twinkling visibility and signal behavior across modalities.

- Conducting wave-physics simulations to model acoustic propagation and scattering using algorithms such as reverse time migration (RTM), full-wave inversion (FWI), phase aberration correction, and total focusing method (TFM).
- Collaborating on preclinical animal studies and human imaging trials, overseeing data collection, image analysis, and experimental design under IACUC and IRB protocols.
- Integrating computational modeling with experimental validation to investigate acoustic field behavior, resonance effects, and artifact generation mechanisms.

RESEARCH EXPERIENCE

“Development of a theoretical foundation for new elastography (elasticity imaging) methods based on acoustic surface wave (Ph.D. dissertation)”

- Investigating surface wave elasticity imaging approach based on Scholte wave using Vantage 256™ on a gelatin phantom
- Quantifying the surface elastic properties of the tissue based on guided wave ultrasound elastography technique
- Simulating Scholte wave elastography using K-Wave and analyzing the different ultrasound beamforming parameters on resolution and sensitivity
- Performing experiments related to the diagnosis of skin cancer using surface wave elastography

“Characterizing the anisotropic properties of the bone scaffolds using Resonance Ultrasound Spectroscopy (RUS)”

- Calculating the eigenvalue solutions and resonance frequencies of a three-dimensional vibrating lattice using analytical approach
- Estimating the natural frequencies of the lattices experimentally and with FEA tool
- Applying RUS technique to retrieve the anisotropic coefficients of the lattices using experimentally measured natural frequencies

“Design and testing of 3D printed tissue scaffolds with directionally tunable stiffness”

- Investigated the lattice properties of height adjustable BC-Tetra structures with ~70% and ~50% porosity
- Designed and tested of a scaffold implant to determine if stiffness is suitable for vertebral fusion
- Demonstrated the mechanical feasibility of BC-Tetra unit cell lattices for spinal fusion devices, with improved multi-axial mechanics compared to conventional cubic unit cell lattices

TEACHING

Course	Term	Role	Number of Students
Thermodynamics-II (ME 3322)	Spring 2020 – Spring 2023 (7 terms)	Instructor/TA	~ 590 (20 sections)
Thermodynamics-I (ME 2322)	Spring 2018 – Fall 2019 (3 terms)	TA	~ 340 (9 sections)
Thermal-Fluid Laboratory (ME 4234)	Summer 2022 – Summer 2023	Instructor	40 (4 sections)
Control of Dynamic Systems / Lab (ME 3360/4234)	Fall 2017, Summer 2022	Instructor/TA	~ 110 (4 sections)
Materials and Mechanics Lab (ME 4217)	Spring 2018	TA	67 (2 sections)

WORK EXPERIENCE

Texas Tech University, Lubbock, Texas, USA	Graduate Research Assistant	Aug 2017 – May 2024
Tesla Inc., Nevada, USA	Intern, Material Flow and Automation	Jan 2023 – May 2023
Dird Group, Dhaka, Bangladesh	Mechanical Engineer	May 2016 – Jul 2017
Coats-Plc, Chittagong, Bangladesh	Mechanical Engineer Intern	Jan 2016 – May 2016

Technical Expertise

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- **Programming & Computational Methods:** C, C++, MATLAB, Python, R, PLC programming, statistical modeling, and numerical optimization.
 - **Modeling & Simulation:** Finite element and wave propagation modeling using ANSYS, Abaqus, FEBio, Simulink, Focus, and Field II.
 - **Signal & Image Processing:** Ultrasound RF-to-IQ conversion, delay-and-sum beamforming, Doppler velocity estimation, twinkling artifact analysis, displacement and strain mapping, spectral analysis, and quantitative image reconstruction.
 - **Experimental Systems & Instrumentation:** Verasonics ultrasound system, optical coherence tomography (OCT), digital image correlation (DIC), function generator, oscilloscope, 3D printing electrospinning.
 - **Biomedical & Material Characterization:** Phased Array Ultrasound Testing (PAUT), ultrasound elastography, Total Focusing Method (TFM), Reverse Time Migration (RTM), Full Waveform Inversion (FWI), tensile and compressive testing.
 - **Computer-Aided Design & Visualization:** SolidWorks, Autodesk Inventor, GD&T, Visual3D, Blender, Adobe Illustrator, and 3D Slicer.
 - **Clinical & Translational Research:** Human and animal study data acquisition and analysis; experienced with Institutional Review Board (IRB) and Institutional Animal Care and Use Committee (IACUC) protocols; ultrasound-guided imaging studies for diagnostic validation and biomarker development.

SELECTED PUBLICATIONS

Journals (in Preparation)

- Mara A. Piltin, Gina K. Hesley, Matthew W. Urban, Blake A. Kassmeyer, **Al Masud, A.**, Nicholas B. Larson, Christine U. Lee. Ultrasound Conspicuity, Safety, and Migration of a Nonmetallic Ultrasound Twinkling Marker Observed for Sonographic Targeting (UTMOST Trial)—Results of a Phase 1 Clinical Trial.
- **Al Masud, A.**, Nwachukwu, C. T., Langenfield, I. D., Trzasko, J. D., Urban, M. W., Lee, C. U. Evaluation of commercial breast biopsy markers with zero echo time MRI (*Submitted to Journal of Magnetic Resonance Imaging*).
- **Al Masud, A.**, Chartier, S. R., Nwachukwu, C. T., Hesley, G. K., Giri, S., Larson, N. B., Langenfield, I. D., Trzasko, J. D., Urban, M. W., Lee, C. U. Evaluation of a polymethyl methacrylate breast biopsy marker with zero echo time MRI. *In preparation for submission to Radiology (Technical Development)*.
- **Masud, A. A.**, Wood, B.G., Lee, C.U. & Urban, M.W., Quantitative Analysis of Ultrasound Doppler Twinkling Artifacts: Spectro-Temporal and Signal Processing Perspectives

Journals (Published)

- **Al Masud, A.**, Egan, P. F., Liu, J., & Fisher, K. A. (2025). A stochastic approach for calculating elastic constants of polymer lattice structures based on spectral ultrasonic data. *Ultrasonics*, 107870.
- **Masud, A. A.**, & Liu, J. (2024). Numerical simulation of impulse-induced surface acoustic waves for elastography purposes using k-Wave simulation toolbox. *Journal of Applied Physics*, 136(16).
- **Al Masud, A.**, Arefin, A., Khatri, N. R., Habib, A. A., Chyu, M. C., & Egan, P. F. (2025). Mechanical Design of 3D Printed Bone Tissue Scaffolds with Tunable Anisotropy. *Biomedical Engineering Advances*, 100170.
- **Masud, A. A.**, & Liu, J. (2024). Ultrasonic surface acoustic wave elastography: A review of basic theories, technical developments, and medical applications. *Medical Physics*.
- **Al Masud, A.**, Shen, C. L., & Chyu, M. C. (2022). On the Optimal Whole-Body Vibration Protocol for Muscle Strength. *Biomechanics*, 2(4), 547-561.
- **Al Masud, A.**, Shen, C. L., Luk, H. Y., & Chyu, M. C. (2022). Impact of Local Vibration Training on Neuromuscular Activity, Muscle Cell, and Muscle Strength: A Review. *Critical Reviews™ in Biomedical Engineering*, 50(1).
- Chumacero, E., **Al Masud, A.**, Isik, D., Shen, C. L., & Chyu, M. C. (2018). Advances in Powered Ankle– Foot Prostheses. *Critical Reviews™ in Biomedical Engineering*, 46(3).

Conference

- **Al Masud, A.**, Chartier, S. R., Nwachukwu, C. T., Hesley, G. K., Giri, S., Larson, N. B., Langenfield, I. D., Trzasko, J. D., Urban, M. W., Lee, C. U. Zero Echo Time Imaging May Improve Mri Visualization Of Breast Biopsy Markers. *RSNA, 2026*, Chicago
- Jeffrey E. Johnson, Gina K. Hesley, Blake A. Kassmeyer, Al Masud, Nicholas B. Larson, Matthew W. Urban, Christine U. Lee, Mara A. Piltin, Safety and Conspicuity of a Nonmetallic Ultrasound Twinkling Marker Observed for Sonographic Targeting (UTMOST Trial), *Global Breast Cancer Conference 2026*, South Korea
- **Masud, A. A.**, Wood, B.G., Lee, C.U. & Urban, M.W., (2025) Spectro-temporal Signal Analysis of Twinkling Artifacts from PMMA Markers. 189th Acoustical Society of America meeting, Honolulu, Hawaii, USA
- **Masud, A. A.**, Wood, B.G., Lee, C.U. & Urban, M.W., (2025) Signal Processing Analysis of Twinkling Artifact in Ultrasound Doppler Imaging. 188th Acoustical Society of America meeting, New Orleans, USA
- **Masud, A. A.**, Karl A Fisher, Egan P & Liu, J. (2024) Estimating Effective Elastic Properties of 3D Printed Polymer Solids and Lattice Structures Using Resonant Ultrasound Spectroscopy. 187th Acoustical Society of America meeting
- Fisher J, Rush K, Sierra S, Egan P, **Al Masud A**, Ferguson M. "Comparing Loop and Note Security in Novel Self-Tensioning Suture Material." Clinical Orthopedic Society Annual Meeting 2024. Tampa, Florida. Sep 2024
- **Masud, A. A.**, & Liu, J. (2023). A Scholte wave-based ultrasound elastography method for imaging superficial tissue. Acoustical Society of America meeting, 153, A265-A265.
- **Masud, A. A.**, Arefin, A. M., Chyu, M. C., & Egan, P. F. (2021, November). Design and Testing of 3D Printed Tissue Scaffolds with Directionally Tunable Stiffness. In ASME International Mechanical Engineering Congress and Exposition (Vol. 85598, p. V005T05A009). American Society of Mechanical Engineers

- **Masud, A. A.**, Arefin, A. M., Chyu, M. C., & Egan, P. F. (2023) Fabrication consistency and mechanics of 3d printed bone tissue scaffolds with tunable anisotropic stiffness. International Design Engineering Technical Conferences

Poster Presentation

- S. Chartier, **A. Masud**, C. Nwachukwu, G. Hesley, M. Urban, J. Trzasko, S. Giri, N. Larson, C. Lee. Biopsy Marker MRI Detection: I Once Was Blind, but Now I ZTE. European Society of Radiology, Vienna, Austria, 2026
- **Al Masud**, A., Arefin, A., Khatri, N. R., Habib, A. A., Chyu, M. C., & Egan, P. F. Mechanics of 3D Printed Bone Tissue Scaffolds with Tunable Anisotropy. 2025 BMES Annual Meeting, San Diego, CA, 2025
- **Masud, A. A.**, Wood, B.G., Lee, C.U. & Urban, M.W., Radiofrequency and IQ Signal Analysis of Pressure-Dependent Twinkling Artifacts in PMMA Breast Biopsy Markers. 2025 IEEE International Ultrasonics Symposium (IUS), Utrecht, Netherlands, 2025.
- Mara A. Piltin, Gina K. Hesley, Matthew W. Urban, Blake A. Kassmeyer, **Masud, A. A.**, Nicholas B. Larson, Christine U. Lee, "Safety, Ultrasound Conspicuity and Migration of a Novel Ultrasound Twinkling Marker Observed for Sonographic Targeting (UTMost Trial); Results of a Phase 1 Clinical Trial." 26th ASBrS Annual Meeting, Las Vegas, May 2025
- Nara Tashjian, Christine U. Lee, Gina K. Hesley, Blake A. Kassmeyer, Nicholas B. Larson, **Abdullah Al Masud**, Matthew W. Urban, Mara A. Piltin, "Twinkle, Twinkle Little Marker—A Phase 1 Trial on Ultrasound Twinkling of Breast Biopsy Markers" 34th annual National Consortium of Breast Centers (NCBC) conference, Las Vegas, March 2025.

PROFESSIONAL SERVICE

- Reviewer, Medical Physics
- Judge, Biomedical acoustic poster competition, American Society of Acoustics, 2025
- Member, American Society of Mechanical Engineer (ASME), 2020 – present
- Member, IEEE, 2025-2026
- Member, American Society of Acoustics (ASA), 2023 – present
- Judge, 3-Minute Thesis competition and graduate poster competition Texas Tech University, 2022

SCHOLARSHIP AND AWARDS

- Dean's Office Scholarship, Texas Tech University, Fall 2017 - present
- Graduate Student Research Support Award, Texas Tech University (Spring 2023)
- Doctoral Fellowships, Texas Tech University, Fall 2023 – Spring 2024
- Travel grant, American Society of Acoustics – (184th & 188th meeting)
- Travel grant, Brillouin Summer School, 2024, Wayne State University, MI